



Module 2: Introduction to NLP

AWS Academy Natural
Language Processing

Module overview

Sections

1. NLP and ML
2. Common NLP tasks
3. Walkthrough of an NLP problem
4. NLP architectures

Module objectives

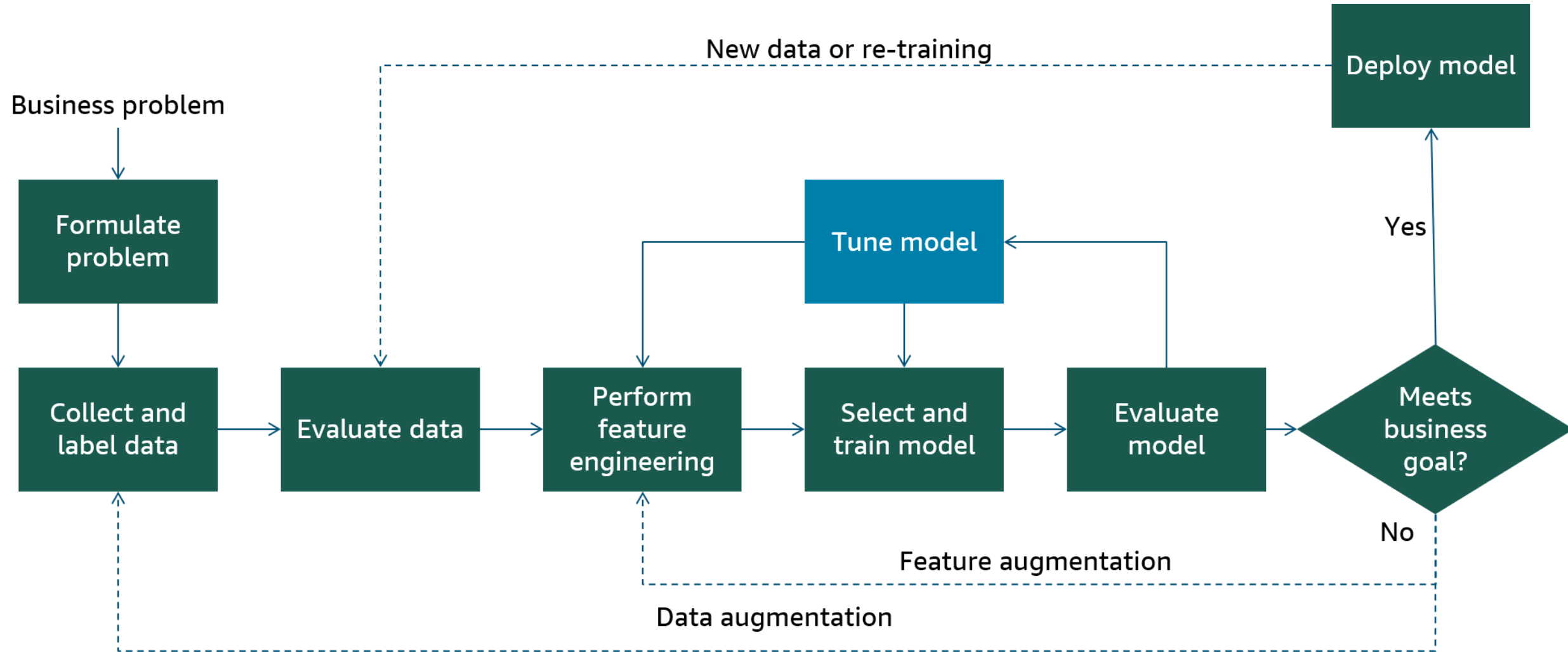
- At the end of this module, you should be able to:
- Describe how the ML pipeline can be applied to NLP
- Identify common NLP tasks
- Use the ML pipeline on an NLP problem
- Summarize NLP architectures



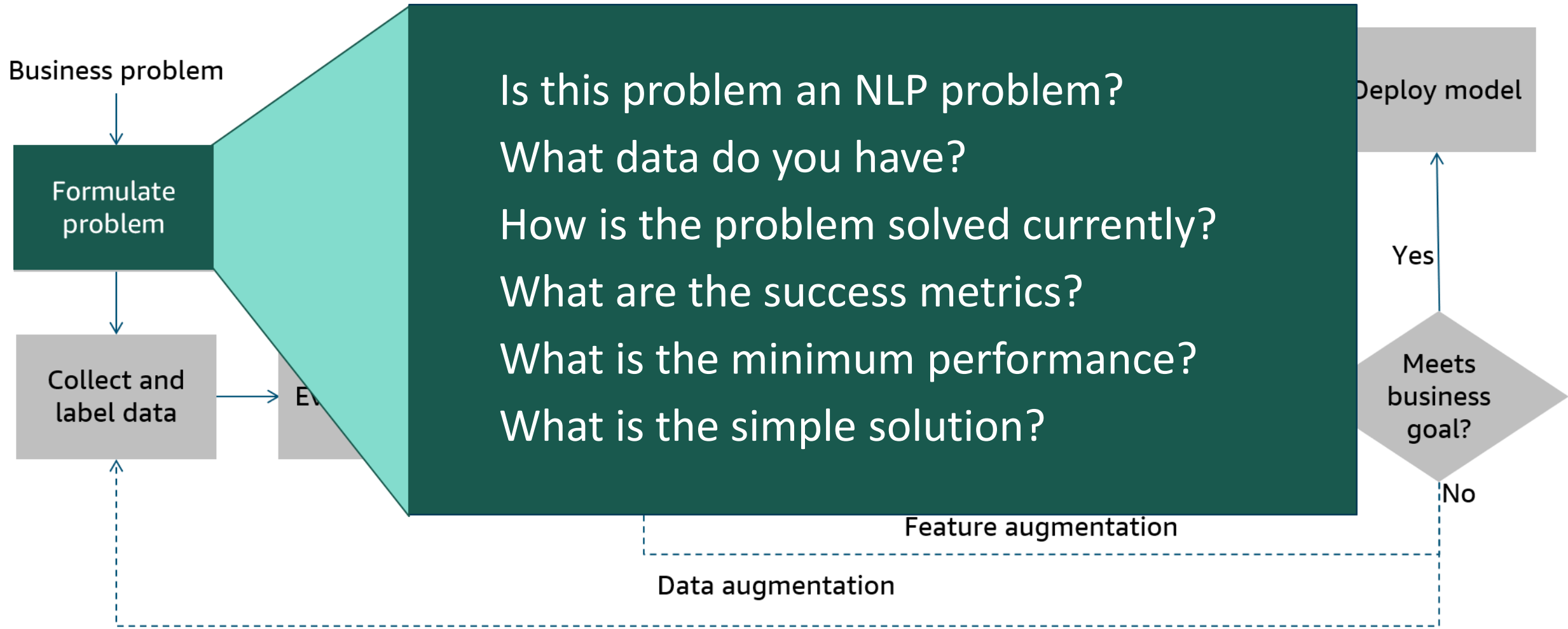
Section 1: NLP and ML

Module 2: Introduction to NLP

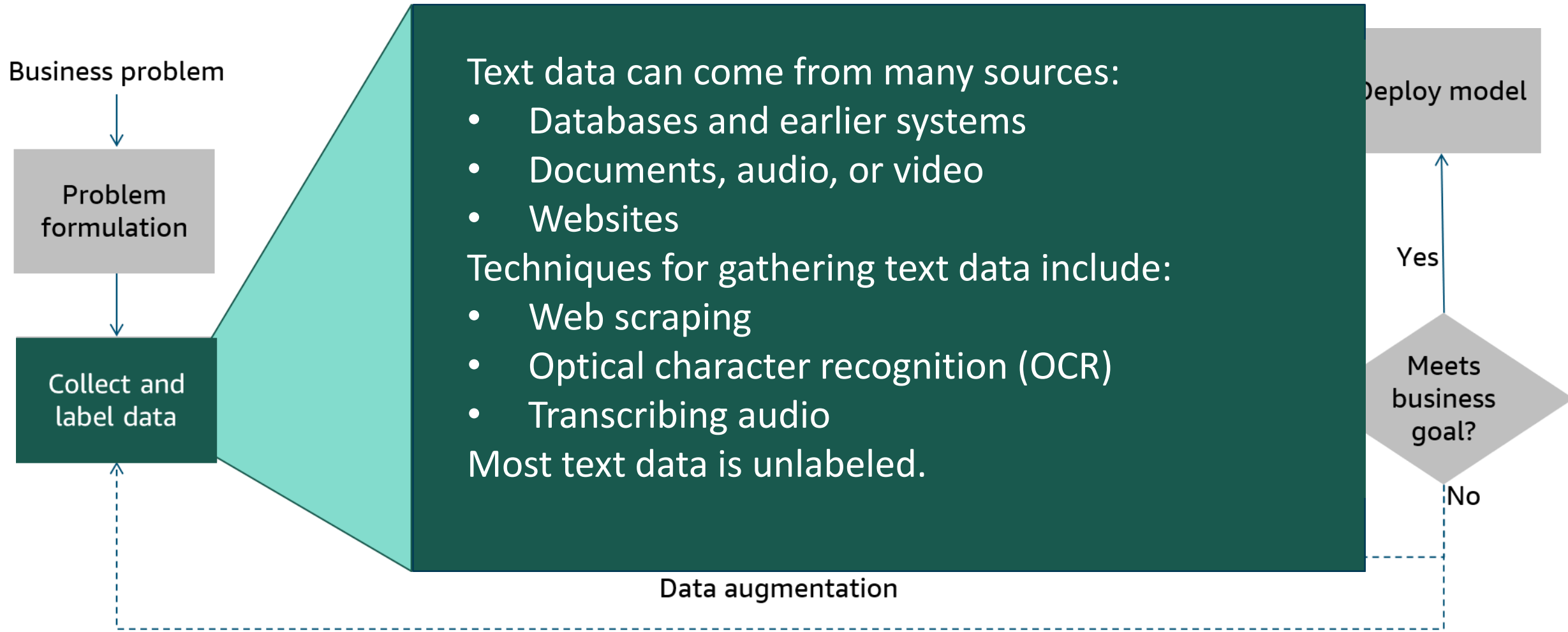
Applying the ML pipeline to NLP



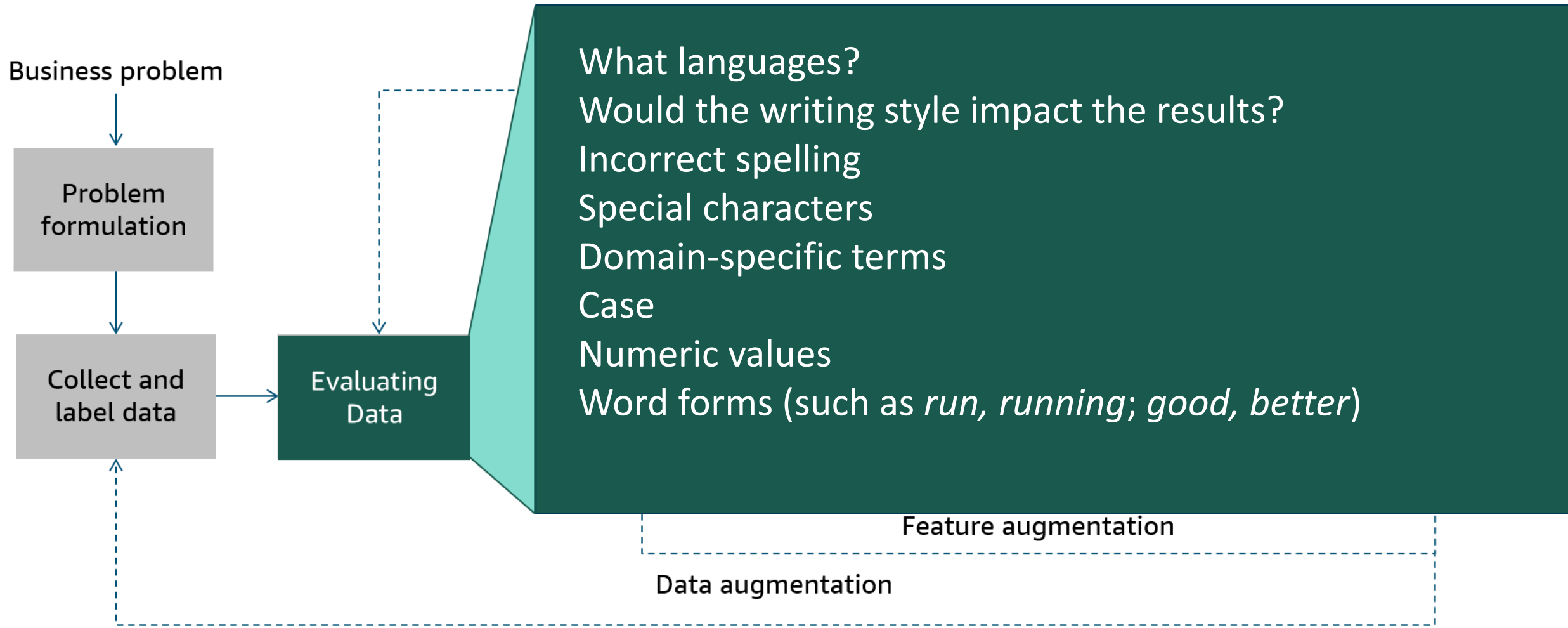
Formulate the NLP Problem



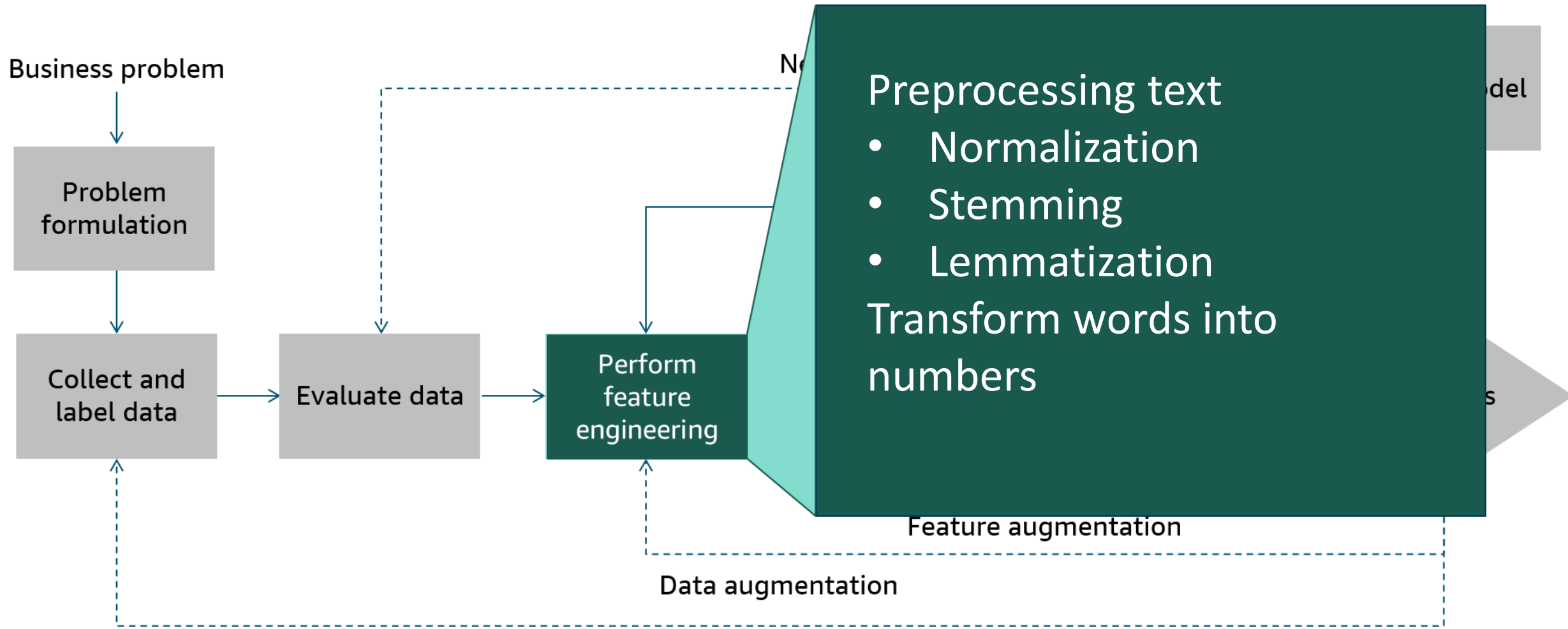
Collecting text data



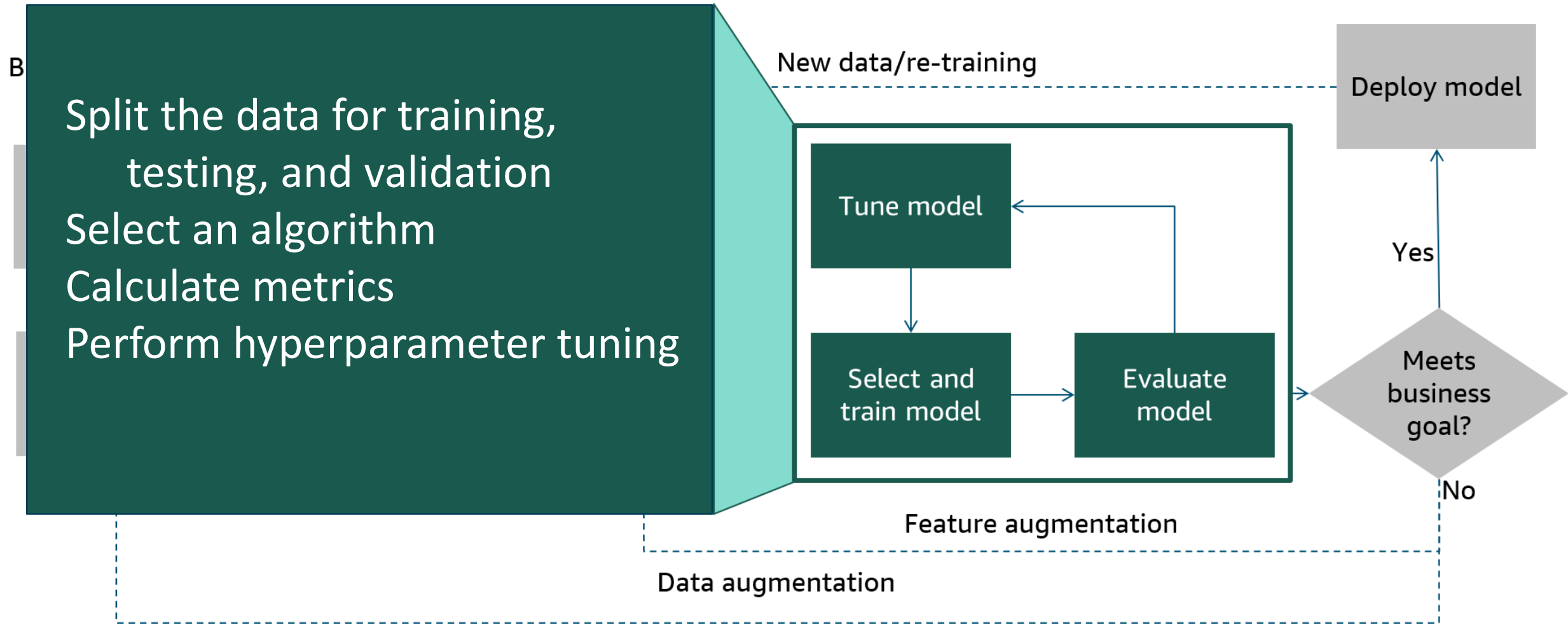
Evaluating text data



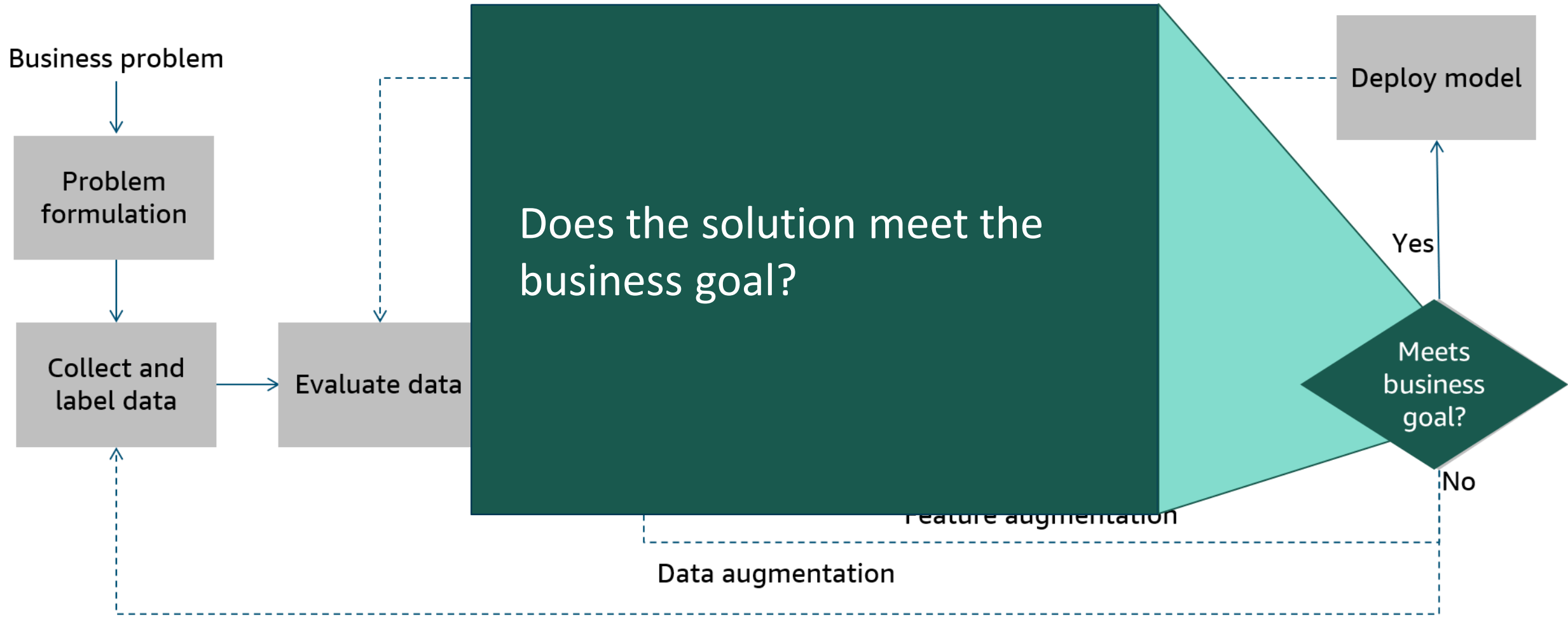
Feature engineering



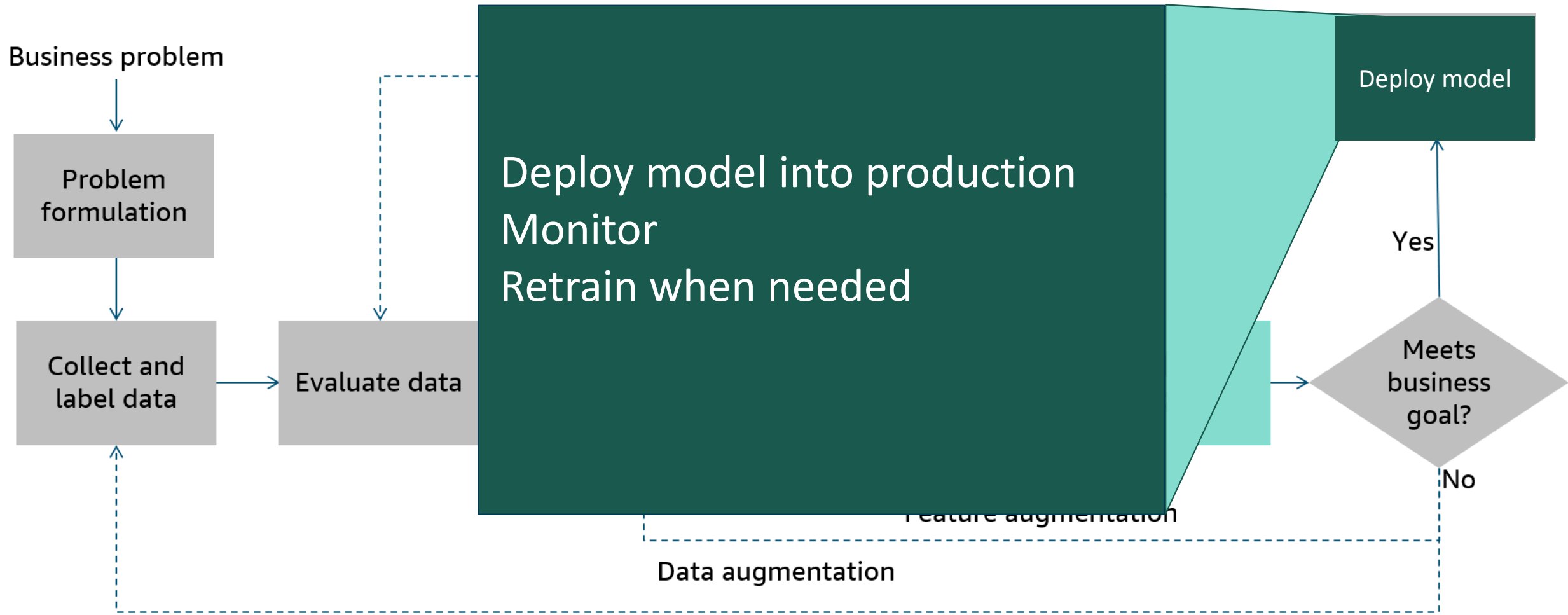
Select, train, evaluate, and tune the model



Does the solution meet the business goal?



Deploy and operate the solution

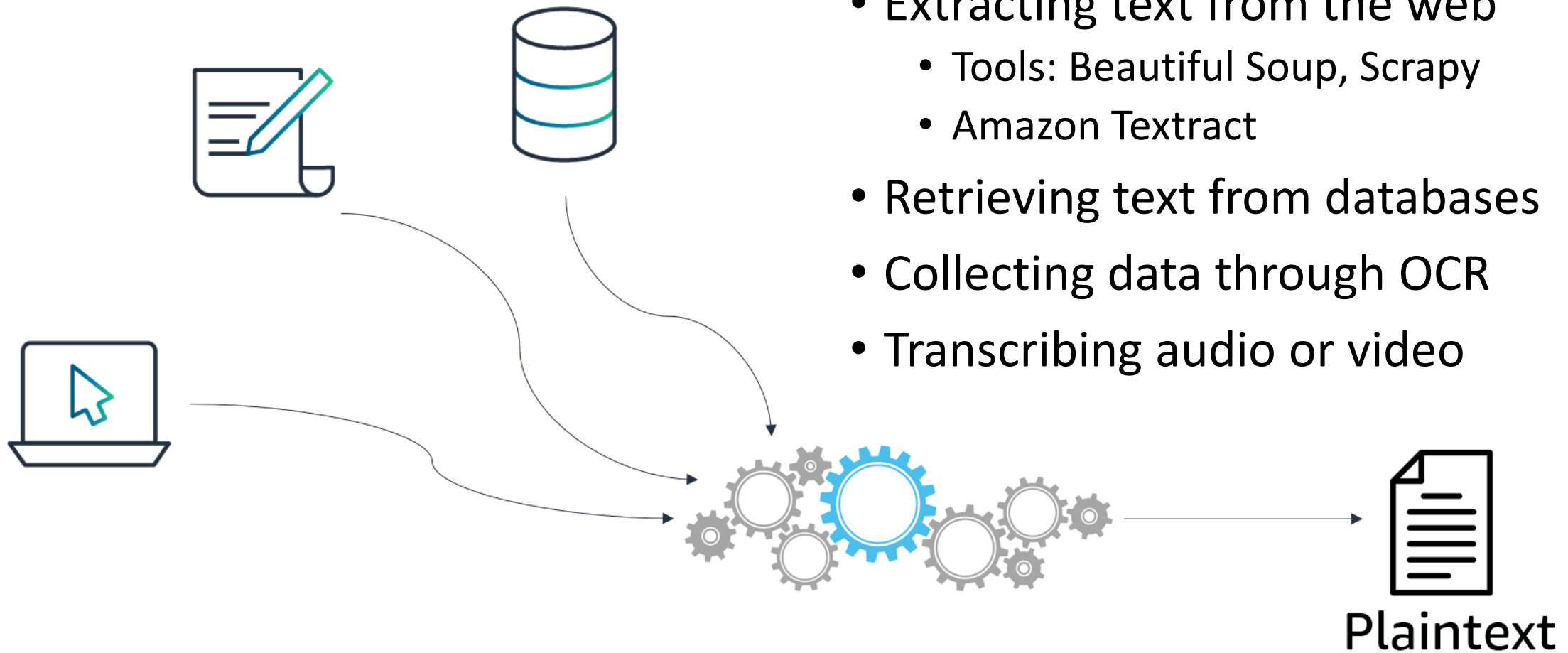




Section 2: Common NLP tasks

Module 2: Introduction to NLP

Tasks for collecting and labeling data



Tasks for text processing

- Removing stopwords
- Performing normalization tasks
 - Removing punctuation
 - Converting numbers to text
 - Converting text to lowercase
- Stemming
- Performing lemmatization
- Tagging parts of speech
- Performing tokenization

Text-processing example

- Task 3: Text preprocessing is an important NLP step b4 training a model. Tasks to correct spelling, converting words to a better standard form & removing STOP words.

Text-processing example: Remove stopwords

- Task 3: Text preprocessing is an important NLP step b4 training a model. Tasks to correct speling, converting words to a better standard form & removing STOPWORDS.

Remove stopwords

- Task 3: Text preprocessing ~~is an~~ important NLP step b4 training ~~a~~ model. Tasks ~~to~~ correct speling, converting words ~~to a~~ better standard form ~~&~~ removing STOPWORDS.
- “Is an”, “to”, “to a”, and the ampersand are removed.

Text-processing example: Normalize

- Task 3: Text preprocessing is an important NLP step b4 training a model. Tasks to correct speling, converting words to a better standard form & removing STOPWORDS.

Remove stopwords

Normalize

- task three text preprocessing important natural language processing step before training model tasks correct spelling converting words better standard form removing stopwords
- All text is now lowercase. NLP acronym is spelled out. “b4” becomes “before”. Punctuation is removed and “speling” is corrected as “spelling”.

Text-processing example: Stemming

- Task 3: Text preprocessing is an important NLP step b4 training a model. Tasks to correct spelling, converting words to a better standard form & removing STOPWORDS.

Remove stopwords

Normalize

Stemming

- task three text preprocessing important natural language processing step before training model tasks correct spelling converting words better standard form removing stopwords
- “-ing” is removed from “preprocessing”, “processing”, “spelling”, “converting”, and “removing”. “-s” is removed from “words”.

Text-processing example: Lemmatization

- Task 3: Text preprocessing is an important NLP step b4 training a model. Tasks to correct spelling, converting words to a better standard form & removing STOPWORDS.

Remove stopwords

Stemming

Normalize

Lemmatization

- task three text preprocess important natural language process **task** before train model task correct spell convert word **good** standard form remove stopwords
- “step” is changed to “task” and “better” is changed to “good”.

Text processing example: End result

- Task 3: Text preprocessing is an important NLP step b4 training a model. Tasks to correct spelling, converting words to a better standard form & removing STOP words.

Remove stopwords

Normalize

Stemming

Lemmatization

task three text preprocess important natural language process task before train model task correct spell convert word good standard form remove stopword

Tokenization: Machines want numbers!

task three text preprocess important natural language process task before
train model task correct spell convert word good standard form remov
stopword

Map text to vectors of numbers



task	three	text	preprocess	important	natural	language	process	before
3	1	1	1	1	1	1	1	1



Section 3: Walkthrough of an NLP problem

Module 2: Introduction to NLP

Business problem:

Is a review positive or negative?

What data do you have?

- Reviews that were labeled positive or negative

How is the problem solved currently?

- Nikki spends time each day reading comments to find negative ones

What are the success metrics?

- Identify all negative reviews so customer service can address them
- It's ok to incorrectly identify positive reviews as negative

What is the minimum performance?

- 90 percent negative reviews detected

Is this an NLP problem?

- It's a binary classification problem!

Obtaining the data

- 70,000 rows in a comma-separated values (CSV) file

Feature	Description
reviewText	Text of the review
summary	Summary of the review
verified	Whether the purchase was verified (True or False)
time	Unix timestamp for the review
log_votes	Logarithm-adjusted votes log (1+votes)
isPositive	Whether the review is positive or negative (1 or 0)

Sample data

	reviewText	summary	verified	time	log_votes	isPositive
0	PURCHASED FOR YOUNGSTER WHO\ nINHERITED MY "TOO...	IDEAL FOR BEGINNER!	True	1361836800	0.000000	1.0
1	unable to open or use	Two Stars	True	1452643200	0.000000	0.0
2	Waste of money!!! It wouldn't load to my system.	Dont buy it!	True	1433289600	0.000000	0.0
3	I attempted to install this OS on two differen...	I attempted to install this OS on two differen...	True	1518912000	0.000000	0.0
4	I've spent 14 fruitless hours over the past tw...	Do NOT Download.	True	1441929600	1.098612	0.0
5	I purchased the home and business because I wa...	Quicken home and business not for amatures	True	1335312000	0.000000	0.0
6	The download doesn't take long at all. And it'...	Great!	True	1377993600	0.000000	1.0
7	This program is positively wonderful for word ...	Terrific for practice.	False	1158364800	2.397895	1.0
8	Fantastic protection!! Great customer support!!	Five Stars	True	1478476800	0.000000	1.0

Exploratory data analysis

What is the distribution of the data?

```
[ ]: df['isPositive'].value_counts()

[ ]: 1.0 43692
      0.0 26308
      Name: isPositive, dtype: int64
```

Do you have missing values?

```
[ ]: df.isna().sum()

[ ]: reviewText      11
      summary       14
      verified       0
      time          0
      log_votes      0
      isPositive     0
      dtype: int64
```

Text processing

Text Samples

PURCHASED FOR YOUNGSTER WHO\nINHERITED MY "TOO
sMALL FOR ME"\nLAPTOP. IDEAL FOR LEARNING A\nFUTURE
GOOD SKILL. HER CHOICE\nOF BOOKS IS A PLUS AS WAS THIS
BOOK!

I've spent 14 fruitless hours over the past two days fruitlessly attempting to install this software on my computer and nothing I've found has worked. I need the software to type proficiently due to disability, and it will not install. The download itself seems to be a corrupted file, I have a fair amount of computer skills and no amount of tinkering has made the program work, so, judging by other reviews, it must be the program itself.

Remove stop words*

Convert text to lowercase

Remove whitespace

Remove extra spaces and tabs

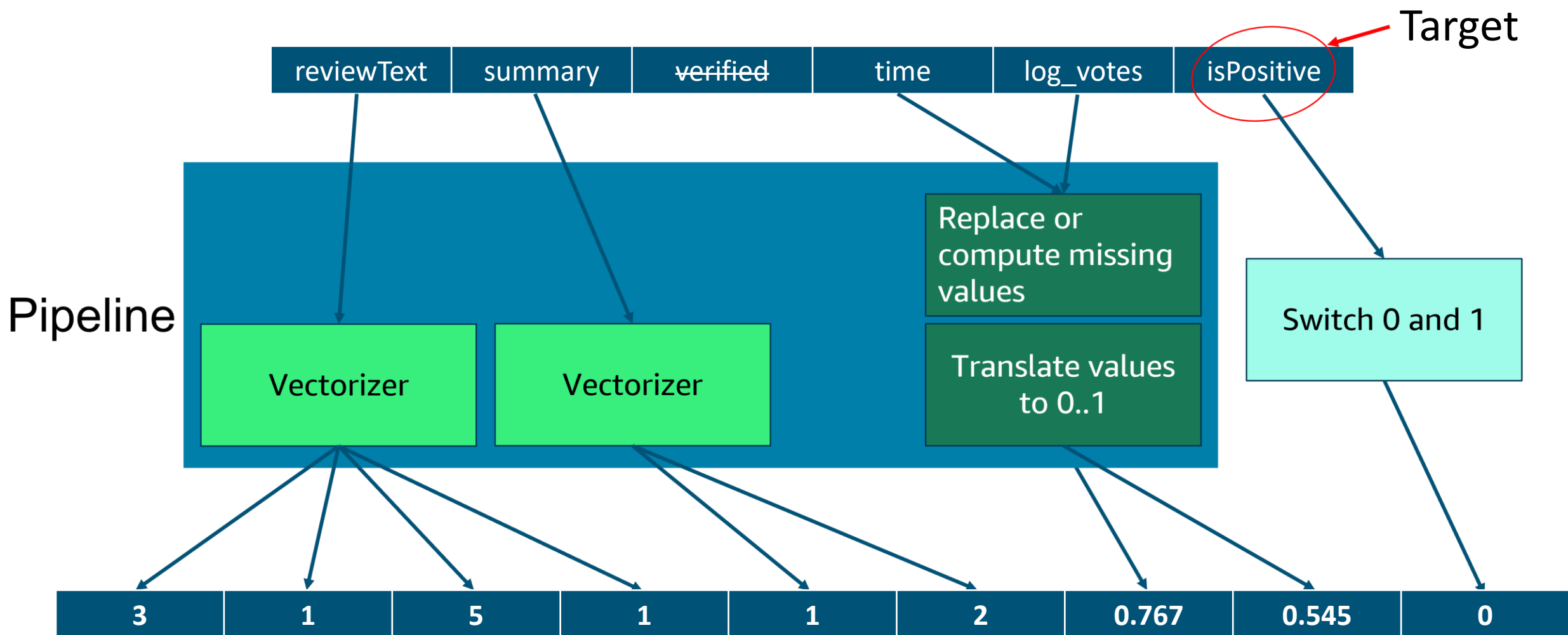
Remove HTML tags

Not removing all stopwords: Explanation

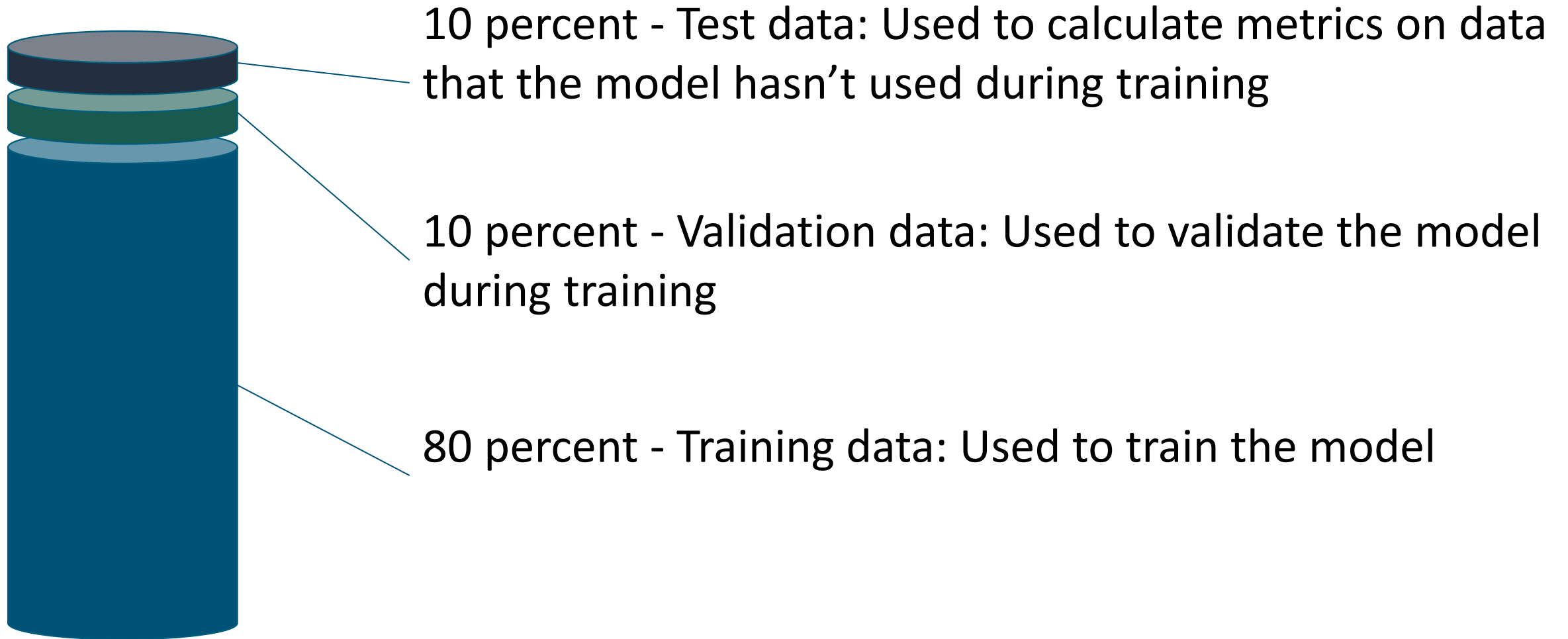
In sentiment analysis, some stopwords are needed!

against, not, do, don't, are, aren't, could, couldn't, did, didn't, does, doesn't, had, hadn't, has, hasn't, have, haven't, is, isn't, might, mightn't, must, mustn't, need, needn't, should, shouldn't, was, wasn't, were, weren't, won, won't, would, wouldn't

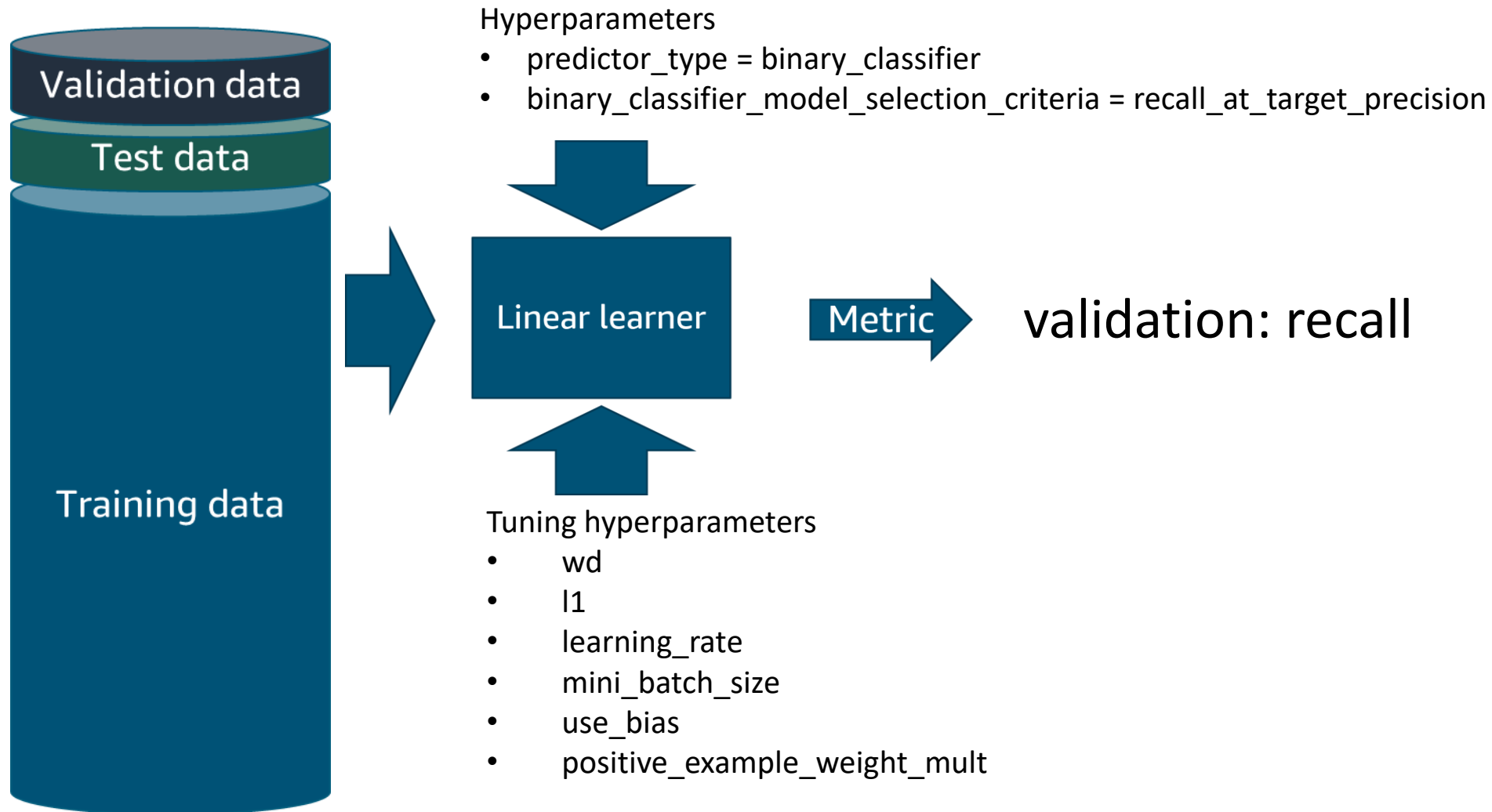
Data processing



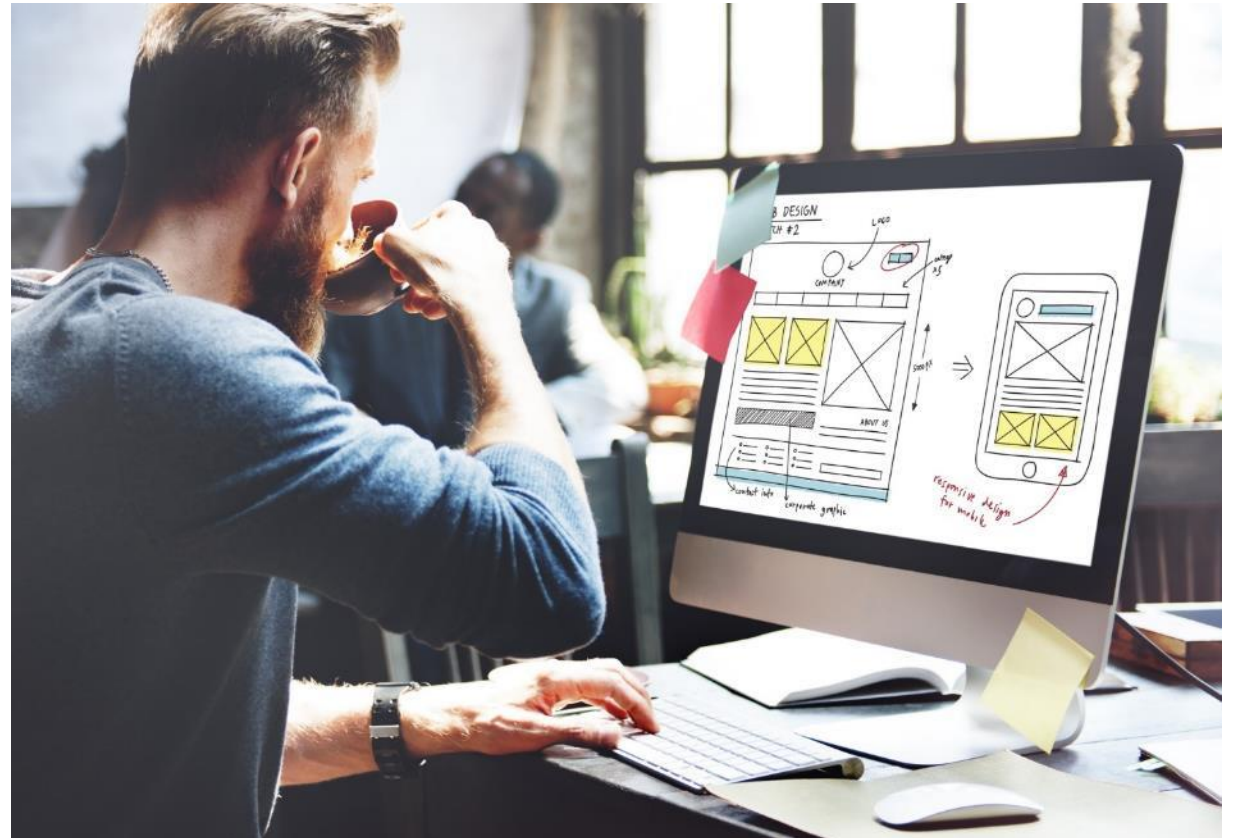
Splitting the data



Training and tuning the model



Module 2 Guided Lab: Applying ML to an NLP Problem

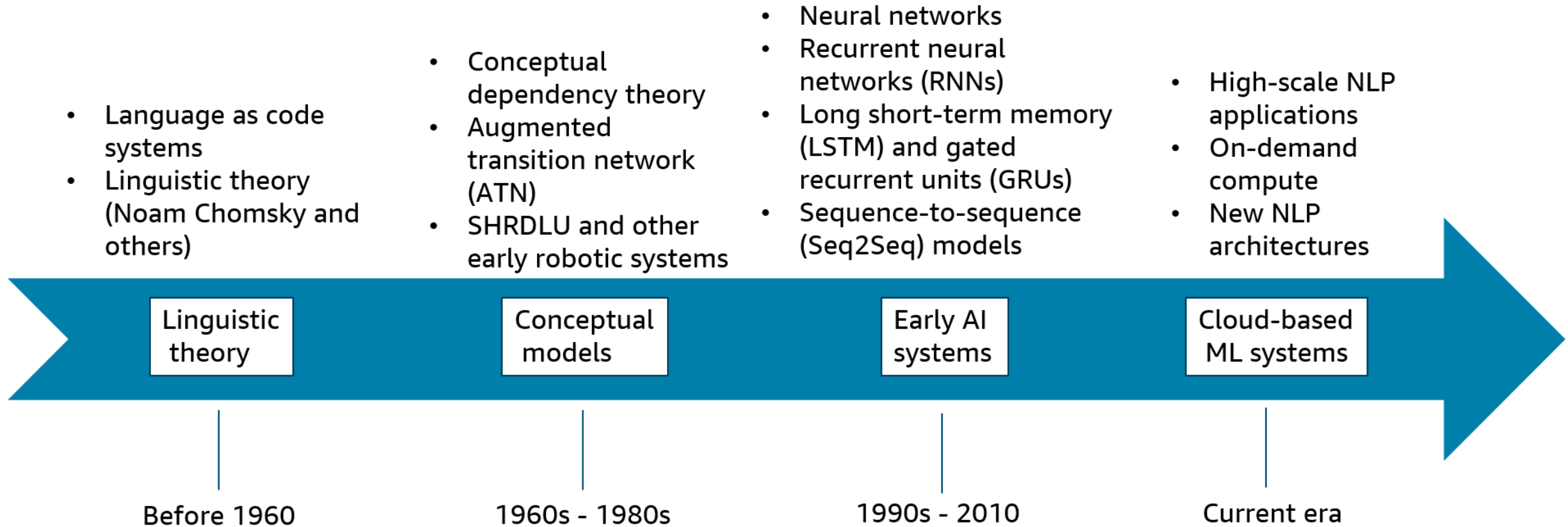




Section 4: Evolution of NLP architectures

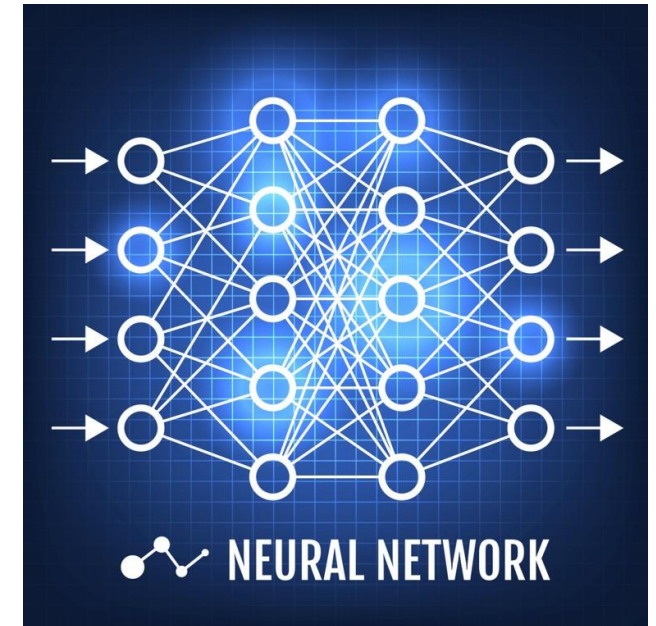
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NLP development timeline



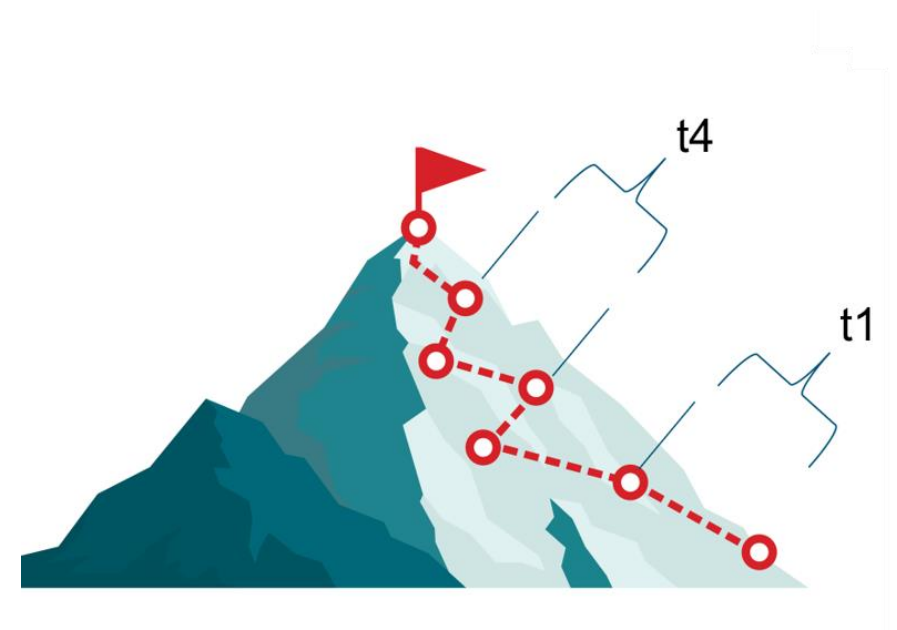
Neural networks

- Neural networks discover relationships in a dataset by adjusting weights for the connections between a collection of nodes
 - Mimics human thinking
 - Is used in unsupervised ML systems
- Generally, neural networks are divided into three layers –
 - Input layer: Captures the dimensions in the dataset
 - Hidden layer: Consists of weighted nodes and activation functions that perform functions specific to the problem domain
 - Output layer: Provides the results of the processing



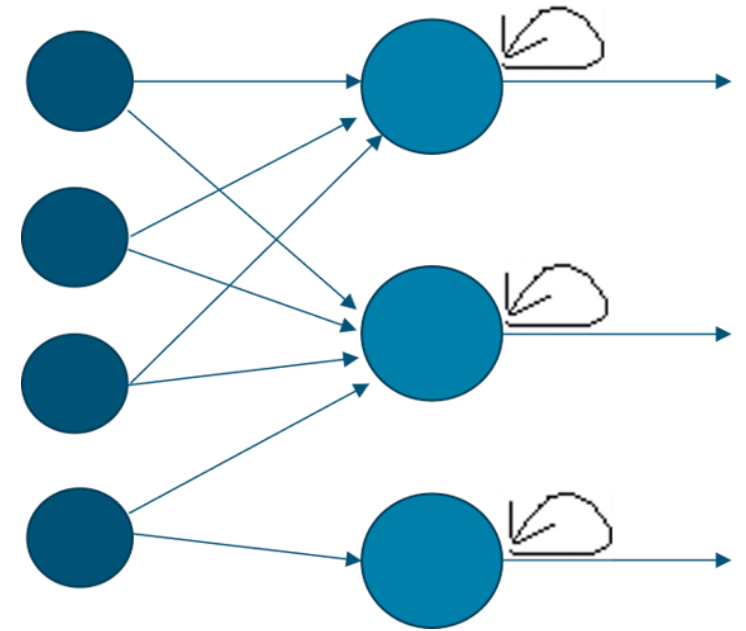
Neural network optimization

- Neural networks must be tuned
 - Find the number of input nodes
 - Determine the best weights for the hidden layer
- Gradients measure the changes to the output relative to the changes in inputs
 - Measures the slope of a function
 - Networks with high gradients learn faster

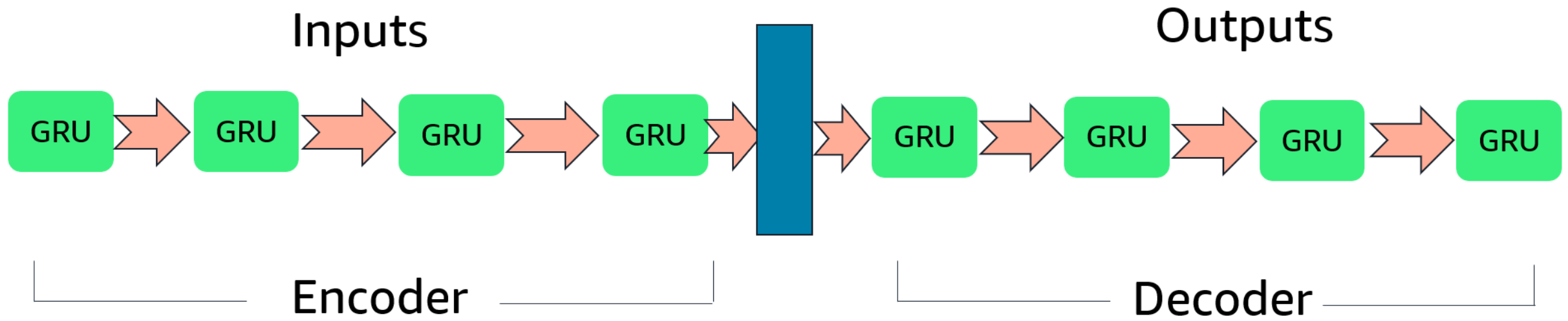


NLP architectures: RNNs

- **Recurrent Neural Networks**
 - Can remember what they learned in previous nodes
 - Are commonly used for NLP problems such as machine translations and chatbots
- RNNs combine short-term memory with *long short-term memory (LSTM)* or *gated recurrent units (GRUs)*



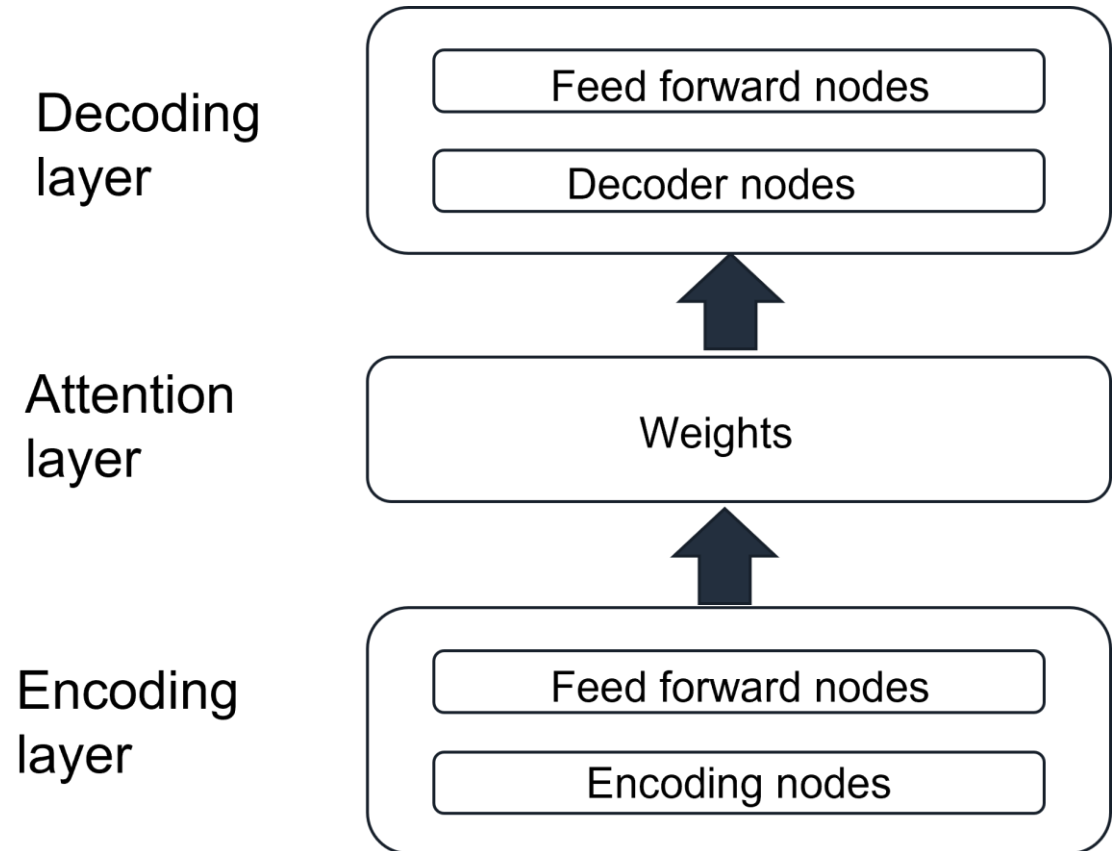
NLP architectures: Seq2Seq models



- Sequence-to-sequence (Seq2Seq) models combine RNNs to solve complex NLP problems
 - For example, the Alexa neural text-to-speech system
- Model consists of:
 - Encoder to collect input
 - Encoder slate or vector
 - Decoder to produce output

NLP architectures: Transformers

- Seq2Seq model + attention mechanism
 - No RNN required
- Attention adds a layer of weights that are used to determine the most important aspects of the input
 - For example, if *order* is important, the attention layer increases the weight for order when it evaluates the input





Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.

Tasks for collecting and labeling data

